Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this `.tex` file directly.

For each question (that applies), replace the "Type your response here" text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace ONLY the "Type your response here" text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference's website to see if you will be asked to provide this checklist with your paper or separately.

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) yes. The Conflict Witness Index subsection gives a conceptual description of the maintained buckets, equivalence classes, constrained-value cells, update path, query path, and shortest-certificate reconstruction.

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) yes. Confirmatory synthetic results, the exploratory public-data topology audit, formal guarantees, limitations, and future work are labeled separately.

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) partial. The paper cites the relevant primary literature on justifications, diagnosis, provenance, incremental view maintenance, sparse and dense retrieval, and rank fusion, but it is not a tutorial survey.

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) yes

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) yes. The store, schema, exact-literal, confidence, time/scope, query relevance, monotone context-cost, bounded-context, and rule-fragment assumptions are stated in the model section and collected again in the supplement.

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) yes. Conflict completeness is defined formally, and detector correctness, joint-budget feasibility and complexity, fixed-hop incompleteness, and the maintained-index contract are proposition statements.

2.4. Proofs of all novel claims are included (yes/partial/no) yes. Core proofs are in the main paper and expanded proofs, including minimality and update-order details, are in the supplementary document.

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) yes. Each proposition is followed by a proof or proof sketch in the main paper, and the supplement provides the expanded argument.

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) yes. Classical justification, diagnosis, provenance, and incremental view-maintenance results are cited where their concepts are used.

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) partial. The propositions are established by proof. Deterministic fixtures and implementation tests exercise their finite executable instances, but experiments do not establish the general theorems.

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) yes. The anonymous code/data archive includes the negative controls, exact comparators, focused tests, and the corrected external-audit script and report.

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) yes. The synthetic fixtures isolate the retrieval contract under an oracle schema, while the separately labeled DBLP–Google Scholar audit tests only whether chained gold-match topology occurs outside the generator.

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) yes. All synthetic fixtures, instance manifests, expected labels, and frozen paper-facing results are included in the code/data archive.

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) partial. The complete reviewer artifact is included with the submission; the public-release license remains an author approval item and is not presumed by the anonymous review package.

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) yes. The DBLP–Google Scholar entity-resolution benchmark is cited in the paper.

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) yes. The external audit uses the public CC-BY release identified by its citation and records the input hashes in the shipped report.

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) NA. No non-public dataset supports a reported result.

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) partial. The final retrieval depths, top-K grid, rank-fusion constant, confidence semantics, fixture families, joint-conflict counts, scale and class-stress sizes, and repetition counts are reported. The manuscript does not reconstruct every development-time trial.

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) yes. Synthetic data generation and the built-in-only external topology audit script are included in the code/data archive.

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) partial. The complete synthetic-fixture, exact, maintained-index, joint-budget, class-stress, on-demand, sparse-retrieval, scoring, and scaling paths are included. The frozen dense outputs and pinned Python environment are included, but reproducing dense embeddings also requires obtaining the cited third-party model weights. For double-blind review, the project-owned RDF namespace is consistently replaced by a reserved `example.org` namespace without changing graph structure or scientific values.

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) partial. Source is available to reviewers now; the public-release license requires author approval before publication.

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) partial. The implementation is typed and tested and documents its invariants and certificate construction, but not every source step has an explicit paper-section cross-reference.

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) yes. Fixtures and outputs have deterministic identifiers or SHA-256 digests, scale-query sampling is deterministic, and the artifact instructions distinguish exact regeneration from measured timing.

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) yes. The paper reports Node.js 22.17.1, Sentence Transformers 5.6.1, the pinned encoder revision, dependency locks, and the AMD Ryzen Threadripper PRO 5955WX timing host with 256 GiB RAM and AlmaLinux 9.7; no GPU was used.

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) yes. Conflict completeness and context cost are defined formally; the evaluation explains why complete minimal-witness inclusion, serialized triple count, joint witness-union cost, update amplification, materialized conflict pairs, and repeated timing are reported.

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) yes. Deterministic fixture and joint-budget results are exact single executions; scale and class-stress timings use one unrecorded warm-up and 11 measured repetitions.

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes. The paper reports per-family means and maxima, exact confusion counts, full family breakdowns, and timing medians with IQRs.

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) no. Deterministic completeness and context results are exact counts, and timing comparisons are descriptive medians and IQRs; the paper makes no statistical-significance claim.

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) yes. The final retrieval depths, top-K grid, rank-fusion constant, encoder revision, rule settings, joint-conflict counts, scale and stress sizes, query counts, and repetition protocol are in the paper and artifact.